HOUSIEN ALSABAG

Stockholm, Sweden • +46736279133 • alsabaghousien@gmail.com • GitHub • LinkedIn • Portfolio

SUMMARY

Computer Engineering graduate from KTH Royal Institute of Technology. Experience across backend development, cloud infrastructure, and AI systems, including Infrastructure as Code on Azure, containerized microservices, RAG pipelines, and multi-agent LLM orchestration. Strong foundation in Java, Python, TypeScript, Spring Boot, Docker, Kubernetes, Terraform, and CI/CD, with practical experience in LLM integration and MCP-based tooling.

WORK EXPERIENCE

Freelance Web Developer, Chef Training & Consultancy (CIC), Dubai, UAE *June 2026 – Present*

- Sole developer across three web properties for a hospitality training company and its affiliated chefs association.
- Built globalarabicchefs.com from scratch in Astro and TypeScript, deployed on Cloudflare Workers.

Thesis Student, Bravura, Stockholm, Sweden *March 2026 – June 2026*

- Designed and implemented reusable Infrastructure as Code (IaC) modules for Azure Functions using Terraform and Bicep, replacing manual resource provisioning with automated, version-controlled deployments.
- Delivered a Terraform and Bicep module library that cut new Azure Function setup from a multi-step manual process to a single pipeline run.
- Authored a comparative thesis report adopted as reference material at Bravura.

Industry Practicum Student (Internship), Hitachi Energy, Sweden *March 2025 – May 2025*

- Designed and implemented a cloud-based monitoring system for electrical substations using Java Spring Boot, Prometheus, and Grafana.
- Deployed containerized microservices architecture using Docker and Azure Container Instances.

EDUCATION

Bachelor of Science in Computer Engineering *August 2023 – June 2026*

KTH Royal Institute of Technology, Stockholm, Sweden

PROJECTS

Marketing Agent Platform | Python, TypeScript/Next.js, PostgreSQL, LiteLLM, Claude, Terraform, Docker

- Building a multi-tenant platform where AI agents decide weekly what to promote, run Meta and Google ads within hard spend limits, and measure results, with human approval on every plan that spends money. Pilot tenant: a Dubai fashion e-commerce brand.
- Designed a Claude Opus orchestrator coordinating six specialist agents through a single gateway that holds all credentials, enforces AI and ad-spend budgets, and logs every external call with one-click undo.
- Next.js dashboard and Python worker on PostgreSQL with per-tenant row-level security; Terraform-provisioned environments, GitHub Actions CI, and 11 architecture decision records.

CodeQ&A, RAG Codebase Assistant | Python, LangChain, ChromaDB, sentence-transformers, Ollama, Streamlit

- Built an end-to-end RAG pipeline that ingests any GitHub repository, chunks source files using language-aware splitters, and embeds them with all-MiniLM-L6-v2 into a persistent ChromaDB vector store.
- Implemented dual retrieval (semantic + filepath-aware) and a local LLM via Ollama to answer natural language questions about codebases with source-grounded responses through a Streamlit UI.

Patient Journal Microservices System | Java, Node.js, React, Docker, Kubernetes, PostgreSQL

- Architected a full-stack, secure medical record application with microservices for patient diagnoses, search, and secure messaging.
- Containerized independent backend services using Docker and orchestrated deployment on KTH Cloud using Kubernetes.

TECHNICAL SKILLS

Languages: Java, Python, Kotlin, C, C#, JavaScript/TypeScript, SQL, Astro
Backend & Cloud: Spring Boot, .NET, Node.js, Next.js, PostgreSQL, Docker, Kubernetes, Azure, AWS, Kafka, Terraform, Bicep
DevOps & Tools: Git, GitHub Actions, CI/CD, Prometheus, Grafana, Keycloak, Postman, Wireshark
AI & Data: LLM integration, multi-agent orchestration, RAG, LiteLLM, LangChain, MCP, PyTorch, scikit-learn, pandas
Certifications: Cisco CCNA: Enterprise Networking, Security & Automation; Switching, Routing & Wireless (2025)

LANGUAGES

English (Native), Arabic (Native), Swedish (Fluent)